

WŌDNESDÆG

SIXTY-FOUR SINES, ONE CHORD

Sixty-four sines, one chord

Wōdnesdæg is a **drone**: sixty-four sines on one chord, each with its own level and its own tuning. Two hands strike them, a sound file runs beside them, and an **EQ**, a reverb and a tape colour the whole.

It comes two ways: **ORIGIN** (the drone, drawn by hand and held on a chord) and **SYNTH** (eight voices, played from a keyboard). Each mode keeps its own sound and its own presets.

Too much to read? Skip it. In ORIGIN, click a few keys on the keyboard, then drag up across the LEVEL row. Open PLAY and let the hands strike it.

In your DAW

The app, the AU (AUv3) and the VST3 are the same instrument. In your DAW you'll find **Wōdnesdæg** under *k.odk*. ORIGIN sounds on its own; MIDI notes hold the chord or, with HAND, strike it. SYNTH plays from MIDI notes.

Installing

1. Open **Wōdnesdæg.pkg** and follow the installer. You can choose: **Standalone + AU**, and/or **VST3**.
2. Open **Wōdnesdæg** once from Applications — this registers the AUv3.
3. In your DAW, rescan plug-ins. Wōdnesdæg is under **k.odk**.

Standalone + AU	The Wōdnesdæg app in /Applications, with its Audio Unit (AUv3) inside: Logic, GarageBand, Live and other AU hosts list it under *k.odk*.
VST3	In /Library/Audio/Plug-Ins/VST3 — Live, Bitwig, Reaper, Cubase …
The app	Plays on its own and follows the Mac's Sound settings: it plays through the output chosen there, and LIVE listens to the input chosen there.

Requires macOS 12 or later · Apple silicon and Intel.

02 · THE SCREEN

The two bars

CHORD VOICE PLAY	The panels: DRONE · AUDIO PROCESS · PLAY (SYNTH: VOICE only). **Done** or Esc closes them.
CLEAR ORIGIN SYNTH	The mode. CLEAR: every LEVEL down and TUNE back · SYNTH: every note off.
MIX GUIDE RANDOM	The MIXER · a short guide (LOG writes a debug log) · RANDOM: a new SYNTH sound.
The sliders	Top — ORIGIN: RANGE, the octaves the 64 spread over · SYNTH: CUTOFF. Bottom — ORIGIN: FILE LEVEL · SYNTH: LEVEL. VOL: the output, its meter under it.

03 · ORIGIN

Level, tune, and the keys



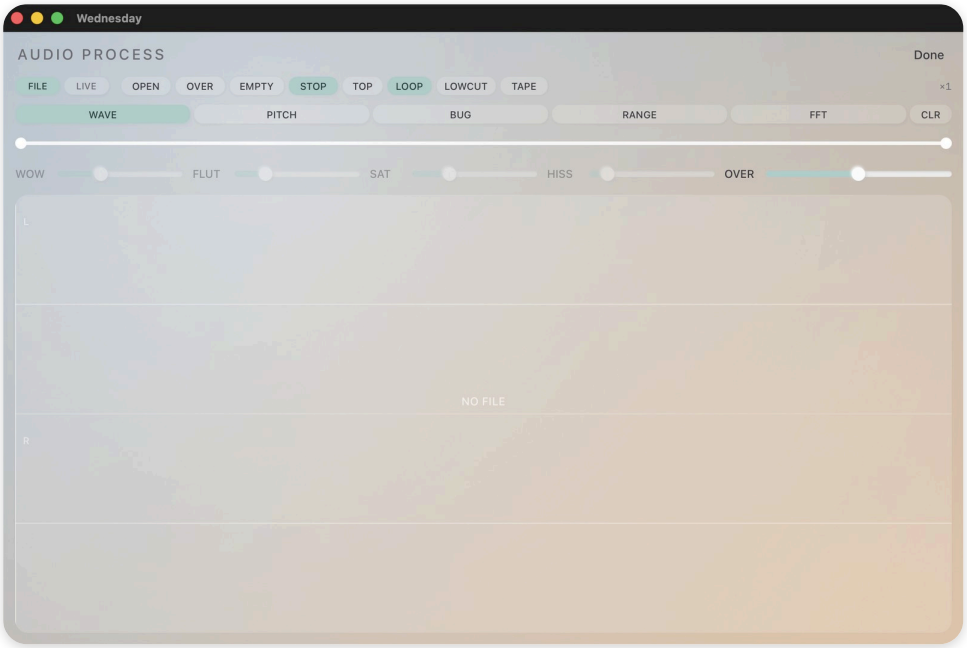
LEVEL	How loud each of the 64 is, low to high. It starts silent: drag across to draw.
TUNE	How far each sine sits off its note: up in mint, down in warm. Drag to draw; double-click puts one back.
The keyboard	Click: hold or let go. The keys held are the chord, the lowest the root; none held, DRONE's KEY and SCALE choose it. MIDI notes hold it too.

Drone



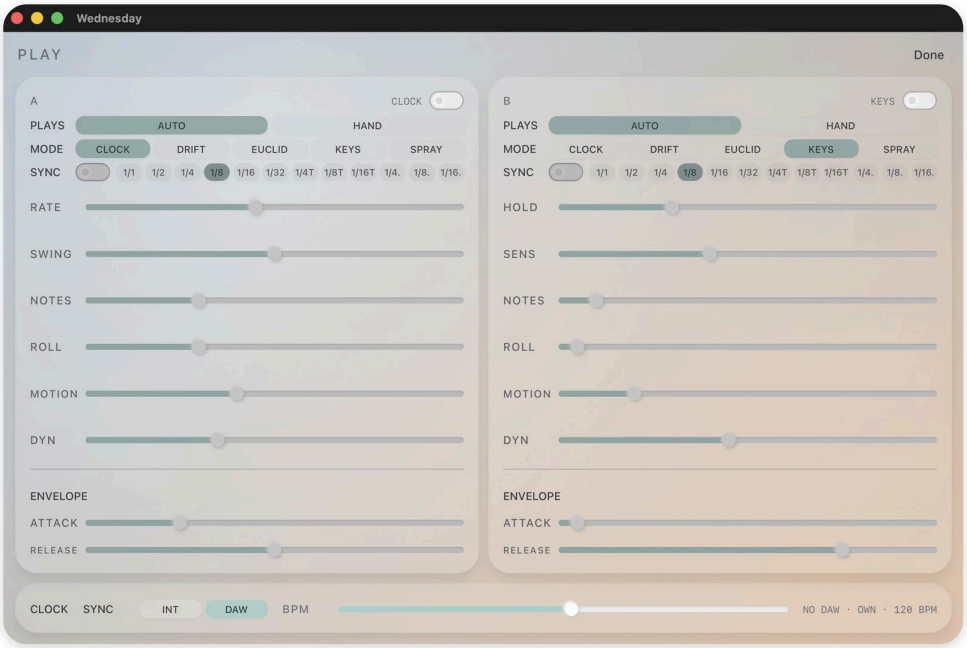
KEY · OCTAVE	The chord's root and its octave (1 … 5).
SCALE · TONES	The temperament's scales, FREE last (KEYS while keys are held). TONES: ALL, or 2 … 7.
MODE	SCALE: the scale over the RANGE · HARM: eight tones, each with eight partials.
TEMPER	JUST · EQUAL · PYTHA · MEAN · SEPT · OVER.
OSC · SHAPE · FOLD	The sines' form: SINE · FOLD · ORGAN · TRI. SHAPE: sine to almost square. FOLD: pushed up and folded back.
GLIDE	How slowly the 64 move to a new chord.
PHASE · INVERT	ALIGN: the 64 start together · SCATTER. INVERT 0 … 3: the lowest tones an octave up.
TIMBRE	The strengths along each eight: FLAT · PURE · ORGAN · TRI · SAW · SQR · BELL.
PAN	MONO · AC (every other) · DC (low and high) · SPAN (low to high), and how wide.
BANDS	Twelve bands over the drone, drawn with a finger. FLAT: all up.
LINK	The file and the drone on each other: FILE (its speed) · DETUNE · LEVEL (its spectrum, THRU BANDS · OSC · VCA) · SHAPE. LIFT: the spectrum lifts the sines.
VOICE	PURE or ANALOG. ALIGN, RETUNE, CLEAR. FROM · TO: the RANGE. KEYS held; LET GO.

Audio process



FILE LIVE	The deck's source: a sound file, or (the app) the Mac's input.
OPEN OVER EMPTY	Open a sound file · lay another over it · empty the deck.
PLAY TOP LOOP	Play and stop · back to the top · loop in the RANGE.
LOWCUT TAPE	The lows under 120 Hz away · the tape on the file.
WOW · FLUT · SAT · HISS	The tape's knobs, when TAPE is on.
OVER	How much a file opened with OVER lays over the one there.
The RANGE bar	Under the pens: where the file loops. The lane below shows L and R.
WAVE PITCH BUG RANGE FFT CLR	What the lane shows and what the pen writes: the sound itself, its pitch, BUG, where it loops, its spectrum. CLR clears the lane.
FILE LEVEL	On the bottom bar: how loud the file is.

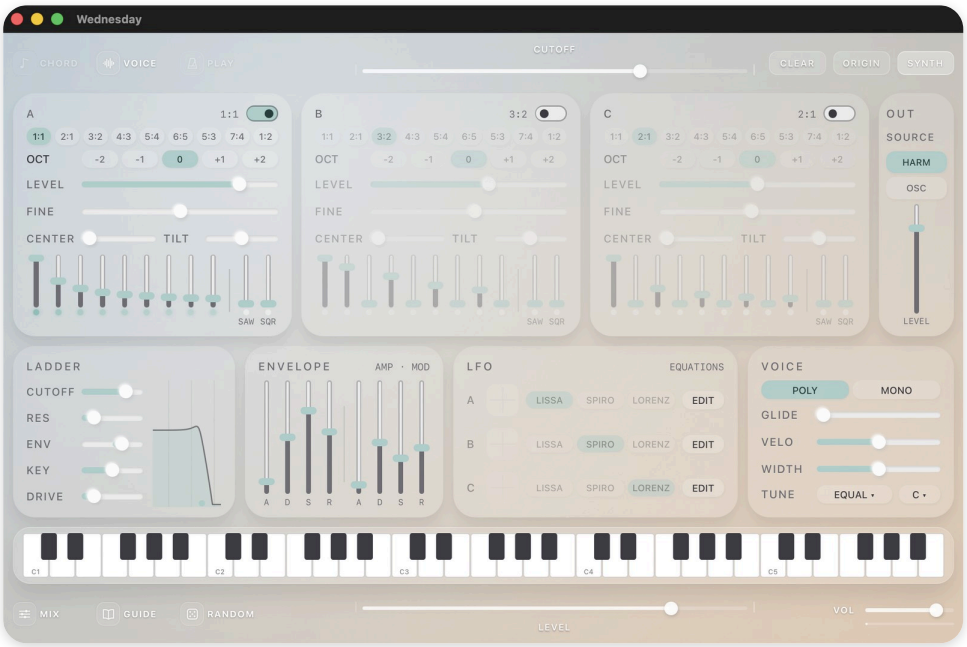
Play



A · B	Two hands striking the 64, each with its own switch.
PLAYS	AUTO: the hand plays by itself · HAND: each key (on screen or from the host) strikes a string at its note, DYN from how hard, the chord kept.
MODE	CLOCK · DRIFT · EUCLID · KEYS · SPRAY — how AUTO strikes.
SYNC	The strikes on the clock's beats: 1/1 … 1/32, triplets and dotted.
RATE · SWING · NOTES	How often · how uneven · how many at once. (KEYS: HOLD and SENS.)
ROLL · MOTION · DYN	Spread in time · how far the strikes wander · how strong.
ENVELOPE	ATTACK and RELEASE of each strike.
CLOCK	SYNC: INT (its own BPM) or DAW (the host's tempo, place and transport; the app alone: its BPM). The LFOs' SYNC follows it too.

Eight voices

Eight voices on the keyboard. OUT chooses the SOURCE: **HARM** (three slots of partials) or **OSC** (one oscillator of 64 sines). A light under each fader shows what sounds now.



HARM OSC	OUT: SOURCE, and the synth's LEVEL.
A · B · C	HARM's three slots, each with its switch. The ratio to the key (1:1 … 1:2), OCT (−2 … +2), LEVEL, FINE.
CENTER · TILT	Where the partials are lifted · their slope.
The faders	The fundamental and eight harmonics, each its own fader, then SAW and SQR.
OSC	One oscillator of 64 sines: OCT, LEVEL, FINE, DETUNE (spreads them). The bars show the 64; SAW, SQR, TRI, PURE, FOLD and ORG mixed.

Filter, envelopes, LFO

LADDER	CUTOFF (also on the top bar), RES (it sings at the top), ENV, KEY, DRIVE.
ENVELOPE	AMP · MOD, each A D S R. MOD opens the ladder by ENV.
VOICE	POLY or MONO, GLIDE, VELO, WIDTH, TUNE (the temperament and its key).

The equation LFOs

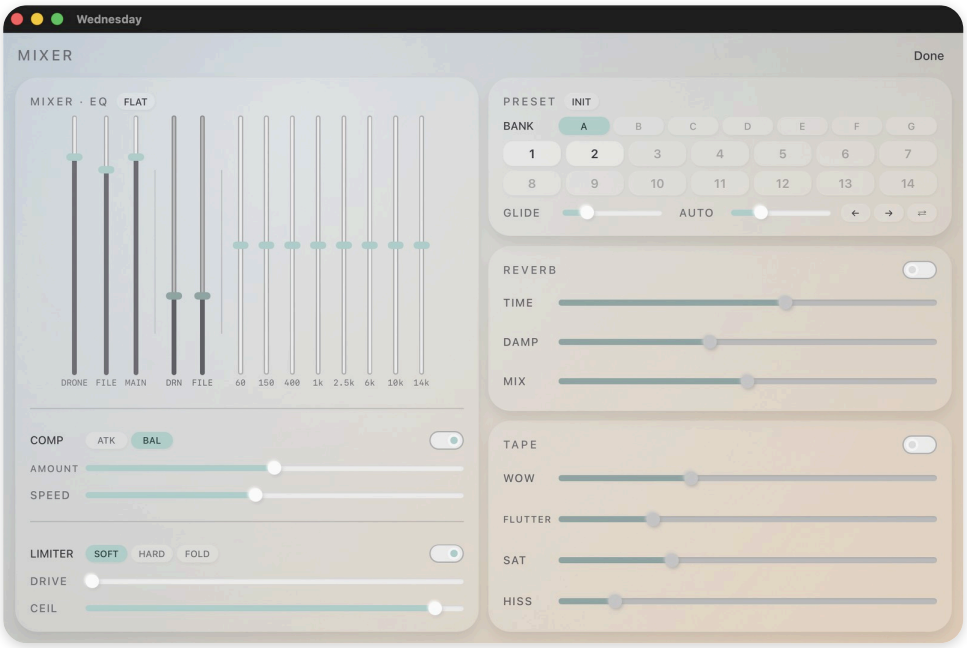
LISSA SPIRO LORENZ	Three figures for A, B and C: a Lissajous, a spirograph, the Lorenz flow. Each traces its figure on the face, dimmed when wired nowhere.
EDIT	SHAPE, RATE, FORM, DEPTH, SYNC (to the CLOCK), and the slots they reach: TO A / B / C.
WIRES	Each out — X · Y · Z · RAD · ANG · SPD — to any places: CENTER, TILT, CUTOFF, RES.
H1 … H9	Nine squares, a partial each in the faders' order: its fader moves up and down with the LFO, the pitch stays; its light shows it.
With OSC	DETUNE, CUTOFF, RES.
RANDOM CLEAR	RANDOM draws wires · CLEAR removes them.

How it is built

Each voice adds its partials (or the 64 sines), runs them through the ladder and the AMP envelope, then out to the MIXER. The equation LFOs move the faders and the ladder from outside. POLY gives each key its own voice; MONO glides from one key to the next. The light under each fader shows what sounds now — the LFOs move it too.

HARM A · B · C / OSC → LADDER → AMP → VOICE → MIXER → out

Levels, EQ and the rest



MIXER · EQ	ORIGIN: DRONE, FILE and MAIN, then DRN and FILE (the sends to the reverb), then the EQ's eight bands (60 … 14k). FLAT: the EQ back. SYNTH: LEVEL and REVERB, then the BANK of twelve.
COMP	ATK (peaks, quick) or BAL (the average, slow), AMOUNT, SPEED. Its switch.
LIMITER	SOFT · HARD · FOLD, DRIVE, CEIL. Its switch.
PRESET	BANK A … G of 14 slots for each mode. Hold a slot: keep · tap: take it back · ⌥-click: empty. INIT. GLIDE: how slowly a preset arrives (up to a minute). AUTO (ORIGIN): the kept presets one after another, ← → or ⇄.
REVERB	TIME, DAMP, MIX — one for both modes. Its switch.
TAPE	WOW, FLUTTER, SAT, HISS. Its switch.

Tips and credits

A first drone	Hold two or three keys, draw LEVEL up in a soft hill, then pull a few TUNE bars a little off: the beating starts.
Wider	Raise RANGE and set PAN to SPAN; TIMBRE BELL brightens the top.
A file	OPEN a field recording in AUDIO PROCESS, then raise LINK's LEVEL: the drone follows its spectrum.
Hands	PLAY A on AUTO, MODE DRIFT, a long RELEASE; PLAY B on HAND for your own strikes.
SYNTH	RANDOM, then wire an LFO to H3 … H9: the partials move by themselves.
In a DAW	CLOCK SYNC DAW: the hands follow the host's tempo and transport.

Credits

Wōdnesdæg — k.odk (Kei Odaka)

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